

Assignment 2

Prediction Electric Engine Status using Audio Analysis and Machine Learning

This assignment will help you develop critical skills in data preprocessing, model selection, and evaluation in classification contexts. It provides an opportunity to gain hands-on experience in addressing real-world machine learning challenges.

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1. Problem Definition

In this assignment, you will use the provided audio dataset to develop models capable of classifying the operational state of an electric engine. The three operational states to classify are: "good," "broken," and "heavyload".

2. Dataset

The dataset, IDMT-ISA-ELECTRIC-ENGINE, was recorded in March 2017 at the Fraunhofer Institute for Digital Media Technology (IDMT) and is referenced in [Grollmisch, Sascha et al., 2019]. It can be downloaded via the link provided in Canvas – Assignment – Assignment 2 page.

As described by [Fraunhofer IDMT, 2017], the IDMT-ISA-ELECTRIC-Engine dataset contains sound recordings from three similar electric engine units (ACT Motor Brushless DC 42BLF01, 4000 RPM, 24VDC), see the figure below.



Figure 1 Electrical engine setup, by [Grollmisch, Sascha et al., 2019].

These recordings simulate different acoustic conditions under three operational states: "good," "broken," and "heavy load." These states were induced by varying the supply voltage and loading weight, which altered the engines' operating sounds. Each recording captures only one engine in one of the three states at a time.

The dataset includes recordings with various background noise types:

- **"pure"**: Recordings with no additional background noise.
- **"talking"**: People talking around the engine casing.
- **"white_noise"**: White noise played through speakers outside the casing.
- **"atmo"**: Atmospheric sounds from a factory environment at three loudness levels ("low," "medium," and "high") played back through speakers.
- **"stress test"**: Simulated setup manipulations, including slightly altered input gains and people knocking on the casing.

The dataset is organized into four folders:

- **"train"**: Original recordings lasting 5-6 minutes each, containing only "pure" recordings.
- **"train-cut"**: 3-seconds segments cut from the original "train" recordings for easier data shuffling and short-term evaluation.
- **"test"**: Original recordings for testing, including different background noise types mentioned above.
- **"test_cut"**: 3-seconds segments cut from the original "test" recordings for easier data shuffling and short-term evaluation.

Each of these folders contains three subfolders: "engine1_good," "engine1_broken," and "engine1_heavyload," where you can find the corresponding recordings, either original or 3-second segments. All recordings are labeled according to their operational state, making them suitable for supervised classification tasks.

Dataset Summary:

Total Recording Duration: 42.32 minutes.

Sampling Rate: 44.1 KHz.

Operational State "good": total 774 samples.

Operational State "broken": total 789 samples.

Operational State "heavy load": total 815 samples.

3. Your Mission

1. **Data Exploration:** Begin by exploring the IDMT-ISA-ELECTRIC-ENGINE dataset. Familiarize yourself with its format, listen to and plot sample recordings, and gain insights into the distribution of samples.
2. **Data Preparation:** conduct necessary preprocessing steps, such as data wrangling, feature extraction, feature scaling, and feature selection. Justify your choices based on the dataset's characteristics.

3. **Classification Model:** Select at least two appropriate classification algorithms and provide a clear explanation of your choice. Utilize the preprocessed training data to train your model, and then evaluate its performance on the preprocessed test data.
4. **Results:** Present the results of your classification model, including accuracy, precision, recall, confusion matrix, and sample predictions. Discuss the model's performance and any notable observations.
5. **Discussion and Conclusion:** Summarize your experiences with the task, highlighting the significance of data preprocessing in improving model performance. Reflect on the challenges faced and insights gained.

4. Submission Requirements

- Prepare a well-structured report that includes sections for data exploration, preprocessing details, model selection, results, and discussion.
- Provide code snippets, visualizations, and explanations to support your work. The original code should be included in the appendix.
- Include your reasoning behind preprocessing choices, algorithm selection, and evaluation metrics selection.
- Offer a thoughtful conclusion summarizing key takeaways.

Good luck, and don't hesitate to seek assistance if needed!

REFERENCES

Grollmisch, S., Abeßer, J., Liebetrau, J., & Lukashevich, H. (2019). IDMT-ISA-Electric-Engine Dataset (1.0.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.7551261>

Grollmisch, S., Abeßer, J., Liebetrau, J., & Lukashevich, H.M. (2019). Sounding Industry: Challenges and Datasets for Industrial Sound Analysis. 2019 27th European Signal Processing Conference (EUSIPCO), 1-5.

Fraunhofer IDMT (2017) IDMT-ISA ELECTRIC ENGINE description, An audio database for industrial sound analysis, version 1.0.

ACKNOWLEDGEMENTS

The original dataset is from Fraunhofer institute for digital media technology IDMT and can be downloaded [here](#).